OmniOmics: a local pipeline hub for reproducible, end-to-end multi-omics analysis with Snakemake and conda

Resource Article draft, target venue: Molecular Ecology Resources

Angelo Felipe Barbosa de Oliveira[1]

1Programa de Pós-Graduação em Genética e Biologia Molecular, Universidade Federal do Pará (UFPA), Belém, PA, Brazil. ORCID: 0000-0003-0831-447X

Draft in progress, started 2026-09-12. All five pipelines' real HPC validation runs completed 2026-09-15 (see github.com/oliv-angelus/OmniOmics-validation for the full results). Sections still marked PENDING (Acknowledgements, Author Contributions, Zenodo DOI, the expanded Table 3/figures) need the author's own input, not further results. This note block itself should be deleted before submission.

Abstract

Wet-lab microbiologists without terminal or HPC fluency are effectively locked out of reproducible multi-omics analysis: workflow managers like Snakemake and Nextflow solve reproducibility but still assume command-line comfort; existing graphical platforms target a single omics type or require dedicated server infrastructure. OmniOmics is a native desktop application wrapping five independent Snakemake pipelines, MAGMA (metagenomics), SIGMA (isolate genomics), DOGMA (metatranscriptomics), DEFINE (differential expression), and PRISM (amplicon metabarcoding), behind one graphical interface with an integrated plotting environment, no server or terminal required. Four of five pipelines were added after the core application with no change to its pipeline execution layer, evidence the architecture generalizes. All five pipelines were validated end to end on real, non-simulated public datasets against independent, external ground truth: certified mock-community compositions, reference genomes, two published papers' own reported results, and a sequencing vendor's own classification of the same read file. Results were strong and mostly concordant, reported honestly where they were not: DOGMA matched 11 of 12 real nitrogen-cycle gene/category directions against Chuckran et al. (2021); MAGMA correctly recovered and classified 7 of 7 metagenome-assembled genomes with no false calls; DEFINE's DEG call rate matched its source paper's order of magnitude with full functional-category overlap; SIGMA's hybrid long-read assembly reached 100% reference genome fraction; PRISM's real-sample taxonomic profiling showed strong compositional agreement (Bray-Curtis 0.08-0.11) with the vendor's own classification of the identical reads. OmniOmics is open source (MIT license), packaged as a .deb, with a companion repository

(github.com/oliv-angelus/OmniOmics-validation) providing every configuration, ground truth, and raw result behind this validation.

1. Introduction

Reproducible multi-omics analysis increasingly depends on workflow managers such as Snakemake (Mölder et al., 2021) or Nextflow (Di Tommaso et al., 2017), which solve reproducibility and dependency management but assume comfort with a command line, a terminal session, and often direct HPC access. This is a real barrier for a large fraction of the researchers who generate the data in the first place, including wet-lab microbiologists, ecologists, and clinicians without dedicated bioinformatics support, who are left either learning enough of the command line to be dangerous, or depending entirely on a collaborator for every new analysis.

Existing solutions address one half of this problem at a time. Galaxy (Galaxy Community, 2022) and KBase (Arkin et al., 2018) provide a graphical interface but require server or cloud infrastructure, whether self-hosted or a shared public instance, rather than a local install a single researcher can simply run. nf-core (Ewels et al., 2020) and Bactopia (Petit & Read, 2020) are local-first and reproducible but remain CLI-anchored; nf-core's own launch command adds a native parameter-entry GUI, but still requires a terminal to invoke and has no run-monitoring or results dashboard without the separate, commercial Seqera Platform layer. Anvi'o (Eren et al., 2021) is local and genuinely multi-omics, but its interactive interface is a downstream visualization layer over already-processed data, not a full configure-run-monitor pipeline execution environment. Bactopia is local-first and easy to run but confined to a single domain, bacterial genomics.

OmniOmics is built to close that specific, compound gap: a desktop-native application, requiring no terminal, no self-hosted server, and no cloud dependency, that covers the entire configure → run → monitor → analyze loop across five independent omics domains in one interface, and is designed so that new pipelines can be added without touching the application's own code.

2. Materials and Methods

2.1 Architecture

OmniOmics is a Flutter desktop application (the graphical client, no HTML/WebView) that spawns a FastAPI/Python backend as a local child process, communicating over HTTP and WebSocket on 127.0.0.1. All pipeline orchestration logic, including conda/Snakemake environment management, database download with progress reporting, Snakemake log parsing, and run monitoring, lives in the backend; the Flutter layer is a thin client.

Each pipeline is a self-contained directory (pipelines/<id>/) holding its own Snakemake workflow and a pipeline.yaml manifest describing its tools, parameters, thread defaults, and database requirements in a declarative schema. The backend discovers installed pipelines by

scanning for this manifest at startup; adding a pipeline means adding a new directory, not modifying application code. Shared databases (e.g., CheckM2, GTDB-Tk, used by more than one pipeline) are deduplicated and configured once at the application level rather than per pipeline.

The five pipelines packaged at the time of writing are: MAGMA, for shotgun metagenomics (assembly, binning, MAG recovery, and annotation); SIGMA, for single bacterial isolate genome assembly, polishing, and annotation; DOGMA, for metatranscriptomics (de novo assembly and functional annotation); DEFINE, for bacterial isolate RNA-seq and differential expression; and PRISM, for amplicon metabarcoding (16S/18S/23S/28S/ITS/rrn-operon/COI/12S, both short-read and long-read).

Beyond running each pipeline, the application includes a Plots tab that generates publication-ready preview figures directly from a completed run's own output (diversity, ordination, differential abundance, and other standard analyses per pipeline), each parametrized through the GUI. This lets a researcher without R programming experience produce and inspect real statistical figures from their own data without writing analysis code.

2.2 Extensibility claim

MAGMA was the founding pipeline; SIGMA, DOGMA, DEFINE, and PRISM were added afterward, each without any change to the pipeline execution layer: discovery, configuration, run submission, and monitoring all worked immediately from the new pipeline's directory and manifest alone. This is presented as direct, structural evidence for the extensibility claim, not an assertion.

The one layer with real, incremental additions per pipeline is the Plots tab described above. Each pipeline's own output shape needed a small, targeted extension the first time it was wired in, for example a new parameter type for SIGMA's per-genome analyses (isolate assembly has no cross-sample table the way a community profile does) and a pipeline-aware output-file lookup for PRISM (a different final-table filename than MAGMA's). These are honestly reported as small, bounded additions to a shared visualization layer, not exceptions to the execution-layer extensibility claim above.

2.3 Comparison to existing platforms (Deliverable A)

Six adjacent platforms were compared on interface, deployment model, domain breadth, and extensibility mechanism, chosen because each occupies a real, citable niche adjacent to OmniOmics's, rather than as a direct tool-for-tool competitor. Comparison used each platform's current public documentation as of 2026-08-27, not only its founding publication's abstract. See Table 1. A second, domain-matched comparison against the single-purpose pipeline a researcher would otherwise use for each of OmniOmics's five domains (Table 1b) makes the same argument at finer grain.

2.4 Real end-to-end validation (Deliverable B)

OmniOmics orchestrates third-party tools without modifying or reimplementing their algorithms; a head-to-head accuracy benchmark against other pipelines would therefore measure those underlying tools, not this contribution's own logic. Instead, each of the five pipelines was run for real, neither as a dry run nor on synthetic data, against a public dataset with an independently known, published expected result, using the identical Snakemake invocation and config schema the application itself generates internally, executed on a compute node (128 CPU cores, ~264 GB RAM, openSUSE Leap 15.3) at Universidade Federal do Oeste do Pará's (UFOPA) institutional cluster. Correctness was then checked against that published ground truth, not against another tool's output.

Datasets, their ground truth, and the pass criterion used for each pipeline are in Table 2. Sample sizes were chosen following standard practice for this kind of infrastructure/orchestration demonstration, rather than for a novel biological replication study, matching nf-core's own minimal test-profile convention and SARTools's (Varet et al., 2016) three-replicates-per-condition validation design, rather than reproducing each source study's full replicate count.

Raw sequencing data is never re-hosted; every dataset is already public (SRA/ENA/GEO/vendor open-data accessions, cited in Table 2). The exact configuration files, curated ground truth, and observed-vs-expected comparison for every pipeline are published as a companion reproducibility package (OmniOmics-validation, see Data Accessibility).

3. Results

3.1 Table 1. Comparison to existing platforms

Platform	Citation	Interface	Deployment	Domain breadth	Extensibility model
Galaxy	Galaxy Community 2022, doi:10.1093/nar/gkac247	Browser (web)	Server/cloud, self-hosted or public	Broad, tool-agnostic	ToolShed XML wrappers, large ecosystem
nf-core	Ewels et al. 2020, doi:10.1038/s41587-020-0439-x	CLI (Nextflow); native parameter-entry GUI (launch), no run-monitoring dashboard without commercial Seqera layer	Local/HPC/ cloud	Broad, community-curated catalog	New pipeline = new community-reviewed repo
KBase	Arkin et al. 2018, doi:10.1038/nbt.4163	Browser (narrative interface)	Cloud-hosted (DOE infra)	Systems/ microbiology/ plant biology	App-based, centrally curated
Anvi'o	Eren et al. 2021,	CLI + interactive	Local	Multi-omics, strongest in	Plugin-style, community-led

	doi:10.1038/s41564-020-00834-3	HTML/D3 visualization, not a run-orchestration GUI		metagenomics	
Bactopia	Petit & Read 2020, doi:10.1128/mSystems.00190-20	CLI (Nextflow)	Local/HPC/ cloud	Single domain: bacterial genomics only	"Bactopia Tools" post-processing modules
Snakemake (engine OmniOmics builds on)	Mölder et al. 2021, doi:10.12688/f1000research.29032.3	CLI, no native GUI	Local/HPC/ cloud	Generic, domain-agnostic	Rules-based, per-project
OmniOmics (this work)	—	Native desktop GUI, zero terminal	Local, zero server/cloud dependency	5 domains in one app	New pipeline = new directory + manifest, no core code change

3.2 Table 1b. Comparison to domain-matched pipelines

Table 1 compares OmniOmics against general multi-tool platforms. A second, more specific comparison is arguably more informative: against the single-domain pipeline a researcher would otherwise use for each of OmniOmics's five domains. All six domain-matched pipelines below are CLI-only (Nextflow-based nf-core pipelines inherit nf-core's own interface model: a native parameter-entry GUI, launch, exists but still requires a terminal to invoke and has no run-monitoring/results dashboard without the separate, commercial Seqera Platform). QIIME2 is the one exception that attempted a native desktop GUI (q2studio), which was never more than an alpha-stage prototype and has been archived since December 2023, unmaintained and unusable with current QIIME2 releases.

OmniOmics pipeline	**Domain-matched tool**	**Citation**	**Native GUI**	**Reference databases**	**Domain scope**
MAGMA	nf-core/mag + nf-core/taxprofiler (two separate pipelines chained together)	Krakau et al. 2022, doi:10.1093/nargab/lqac007; Stamouli et al. 2023 (preprint), doi:10.1101/2023.10.20.563221	None	Auto-downloads GTDB-Tk (~110 GB) and CAT/NCBI-nr (~200 GB) as two separate standalone downloads	mag: assembly/binning/MAG taxonomy only, no read-level community profiling; taxprofiler: read-level profiling only, no assembly
SIGMA	nf-core/bacass	nf-core community, github.com/nf-core/bacass	None	Kraken2/ KmerFinder database paths must be supplied	Bacterial isolate assembly only

				manually	
DOGMA	nf-core/ metatdenovo	Di Leo et al. 2025, doi:10.7717/peerj.20328	None	Per-database manual configuration	Metatranscriptome assembly/annotation only
DEFINE	nf-core/rnaseq + nf-core/differentialabundance (two separate pipelines chained together)	nf-core community	None	Reference genome/annotation supplied by the user	Differential expression only
PRISM	nf-core/ampliseq	nf-core community, github.com/nf-core/ampliseq	None	Auto-downloads SILVA/UNITE/PR2/GTDB on demand	Amplicon metabarcoding only
PRISM (alt.)	QIIME2	Bolyen et al. 2019, doi:10.1038/s41587-019-0209-9	q2studio: archived Dec. 2023, alpha-stage prototype, unmaintained	Manual per-plugin configuration	Amplicon metabarcoding only

Reproducing what OmniOmics covers in one application would require installing, learning, and separately running at least seven different Nextflow pipelines, none of which share a common interface, database manager, or run-monitoring view with each other. Reference-database handling in the nf-core ecosystem is itself real and often well engineered (nf-core/mag and nf-core/ampliseq both auto-download curated databases), which is why this is not presented as a database-automation gap; the actual gap is that database management stays siloed per pipeline, with no shared, GUI-driven view across domains, the same gap that applies to configuration, execution, and monitoring throughout this comparison.

3.3 Table 2. Validation datasets, ground truth, and pass criterion

Pipeline	Dataset	Ground truth	Pass criterion
MAGMA	ZymoBIOMICS mock community, even+log, hybrid Illumina+ONT (Nicholls et al. 2019, doi:10.1093/gigascience/giz043)	Certified theoretical composition (%) per species	Observed abundance within 10% of certified value; correct species-level taxonomy for all 10
SIGMA	Individual E. coli isolate, same Zymo series, Illumina + independent ONT/PacBio (McIntyre et al. 2019, doi:10.1038/s41467-019-08289-9)	Zymo's own reference genome for the isolate	QUAST vs. reference: genome fraction, N50, misassemblies; hybrid vs. short-only comparison
DOGMA	Soil metatranscriptome, glucose-amendment time course (Chuckran et al. 2021, doi:10.1128/mSystems.00161-21)	Direction of change (up/down) for named gene categories at t8 vs t0, real fold-changes from the paper	DESeq2 log2FoldChange sign matches reported direction
DEFINE	E. coli exposed to 239Pu vs. control, day 1 (GEO	590 DEGs (13.8% of CDSs) reported for this	DEG count same order of magnitude; enriched

	GSE208658, doi:10.1128/msystems.00718-22)	contrast; enriched functional categories	categories overlap
PRISM	ZymoBIOMICS 16S “MSPlus” community, ONT (vendor open-data, zymo_16s_2025.09)	ONT's own official Emu classification for the same read file	Species-level abundance correlates with ONT's own result; all 4 spiked species detected

3.4 Real validation outcomes

Pipeline	Status	Headline result
SIGMA (ONT path)	Complete, PASS	Hybrid Illumina+ONT assembly of a Zymo reference E. coli isolate recovered 100.000% genome fraction against the certified reference genome, zero misassemblies, total length within 11 bp of the reference (4,875,452 vs. 4,875,441 bp). Short-read-only assembly of the same isolate, correctly more fragmented (87 aligned contigs vs. 2) but still zero misassemblies, demonstrating measurable improvement from long-read polishing without introducing structural error.
SIGMA (PacBio path)	Complete, mixed outcome (see Discussion)	Hybrid Illumina+PacBio assembly of the same isolate (PacBio RS II CLR subreads, McIntyre et al. 2019) correctly identified the species (GTDB-Tk) with 100.0% CheckM2 completeness and 99.04% genome fraction against the reference, but introduced 21 real misassemblies (6 misassembled contigs, 8 local); the short-read-only assembly of the same isolate had zero. Attributed to the CLR chemistry's substantially higher raw per-read error rate relative to the ONT run above (no circular-consensus correction in this 2018-era dataset), not a pipeline defect; reported as an honest, mixed result rather than a uniform PASS.
DEFINE	Complete, PASS	Real DESeq2 analysis (E. coli exposed to 239Pu vs. non-irradiated control, day 1, 3 replicates per condition) against the real GEO GSE208658 reference genome, using the paper's own stated DEG criteria (log2FC>2 and padj<0.05, Wintenberg et al. 2023) rather than padj alone, found 433 significant DEGs (9.9% of tested genes), the same order of magnitude as the paper's reported 590 (13.8%). Functional-category overlap with the paper also held: significant genes were real hits in every category reported as enriched for this contrast (ABC transporters, siderophore biosynthesis, stress response, amino acid biosynthesis).
PRISM (ground-truth run, ONT 16S “MSPlus”)	Complete, mixed outcome (see Discussion)	Real Emu classification (GTDB SSU r220 reference) of 3 ONT MSPlus replicates, compared against ONT's own official Emu classification of the same read files, at genus level (species-level nomenclature differs between GTDB releases). Bray-Curtis dissimilarity, the standard compositional-similarity metric for this kind of data, was 0.08-0.11 across replicates (mean 0.10), indicating strong community-level agreement; Pearson r (0.96-0.98) and Spearman rho (0.66-0.79) are reported alongside it rather than in isolation, since Pearson alone is dominated by a few high-abundance genera. Of the four spiked-in species specifically tracked, three were detected with closely matching abundances (Bifidobacterium adolescentis, B. vaginale, Borreliella/Borrelia burgdorferi); Chlamydia trachomatis was not detected at all, most likely due to the documented mismatch between universal 16S primers and Chlamydiae sequences rather than a database or

		classification defect (the reference genome is present in the database used).
MAGMA	Complete, mixed outcome (see Discussion)	Real hybrid Illumina+Nanopore assembly and binning of the ZymoBIOMICS even and log-distributed mock communities. Two independent lines of evidence: (1) read-based taxonomic profiling (Kraken2+Bracken) detected all 10 reference species in both distributions, spanning roughly five orders of magnitude in the log-distributed sample, but matched the certified composition within a pre-registered ±10%-relative tolerance for only 2 of 10 species on the equal-abundance sample; real deviations for the remaining 8 (0.5-3.6 percentage points) are consistent with known extraction-efficiency and genome-copy-number biases in read-based abundance estimation, not a pipeline defect. (2) Real metagenome assembly and binning recovered 7 dereplicated MAGs, all 7 correctly classified to the right species by GTDB-Tk with no false calls and high to medium quality (CheckM2 completeness 81.7-100%, contamination 0.0-5.6%), the pipeline's actual differentiator from simple read-classification tools.
DOGMA	Complete, strong result (see Discussion)	Real metatranscriptome assembly (rnaSPAdes+MEGAHIT+Trinity ensemble, MMseqs2-deduplicated) and functional annotation of 4 real soil metatranscriptome samples (2 replicates x 2 timepoints, glucose amendment, Chuckran et al. 2021). Since each sample is assembled independently (no shared reference transcriptome across samples), differential expression was computed by pooling Salmon's per-transcript read counts onto each sample's own gene-family annotation (eggNOG/NCycDB), then running DESeq2 (C8 vs. C0) on the resulting gene-family count matrix. 11 of 12 detected genes/categories (92%) matched the paper's reported direction of change, including every gene/category the paper itself reports as a strong effect: the ammonium transporter amtB, the GS-GOGAT nitrogen-assimilation pathway, and the nitrogen regulatory network all significantly upregulated at 8h (padj<=1.8e-4); nitrification and denitrification genes both significantly downregulated (padj<=0.04). The one mismatch, crp, a global carbon-catabolite regulator rather than a nitrogen-cycle gene per se, was a subtle, non-significant effect in both this run and the original paper.

4. Discussion

SIGMA's two long-read paths on the same isolate illustrate a point worth stating explicitly rather than only reporting a PASS/FAIL count: assembly quality is a function of input data quality, and a real validation should show that honestly. The ONT path (modern chemistry) improved on every axis over the short-read-only baseline, with fewer contigs, higher N50, and genome fraction reaching 100.000% without introducing a single misassembly. The PacBio path, built from 2018-era RS II CLR subreads with no circular-consensus correction, recovered more of the genome and higher completeness than short-only, but at a real cost: 21 misassemblies the short-read assembly did not have. Neither result was adjusted or selectively reported after the fact; both are the direct QUAST-vs-reference output. Presenting the PacBio result as a mixed outcome rather than reframing it as a pass is itself part of the validation's honesty: it demonstrates that

OmniOmics faithfully reproduces whatever the underlying tools and input data actually produce, rather than the demonstration being tuned toward uniformly clean results.

DEFINE's real validation compared its DESeq2 output against the source paper's own stated criteria for calling a gene differentially expressed (log2 fold-change greater than 2 and padj<0.05; Wintenberg et al. 2023), rather than an adjusted p-value cutoff alone, since the paper's headline DEG count was reported under that two-part definition. Under those matched criteria, DEFINE's real output gave 433 significant genes (9.9% of tested genes) against the paper's reported 590 (13.8%), the same order of magnitude. The functional-category check passed as well: significant genes were real hits in every category the paper reports as enriched for this contrast (ABC transporters, siderophore biosynthesis, stress response, amino acid biosynthesis).

MAGMA's real validation is best read as two separate claims with different outcomes, deliberately reported that way rather than collapsed into one score. Read-based taxonomic profiling alone, the same kind of result nf-core/taxprofiler or a plain Kraken2 run would give, detected all 10 reference species correctly in both the equal-abundance and log-distributed communities, but quantitatively matched the certified composition within tolerance for only 2 of 10 species. That gap is a well-characterized property of read-based abundance estimation on real mock-community sequencing data (DNA extraction efficiency and genome-copy-number effects), not evidence against the orchestration layer. The claim MAGMA actually makes beyond a taxonomic classifier, real genome-resolved recovery via assembly and binning, held up cleanly: 7 dereplicated MAGs, every one correctly classified to species by GTDB-Tk, zero false calls, consistently high completeness and low contamination. Reporting the weaker of the two results (read-based quantitative accuracy) alongside the stronger one (MAG recovery), rather than leading only with whichever looks better, keeps this validation to the same standard applied to SIGMA, DEFINE, and PRISM above.

DOGMA's real validation is the strongest directional agreement of the five pipelines: 11 of 12 detected gene-family targets from Chuckran et al. (2021)'s own reported results matched in sign, including every gene/category the paper itself reports as a strong, high-confidence effect (the ammonium transporter amtB, the GS-GOGAT nitrogen-assimilation pathway, the nitrogen regulatory network, and both nitrification and denitrification, all significant at padj<=0.04). Because DOGMA assembles each sample independently, a deliberate design choice since no single de novo assembler is uniformly best for complex environmental metatranscriptomes (Krinos et al., 2023) and pooling reads across samples before assembly would blur the very t0-vs-t8 transcriptional shift under test, there is no shared reference transcriptome across samples for a conventional gene-level count matrix. Differential expression was instead computed by pooling Salmon's transcript-level counts onto each sample's own gene-family annotation (eggNOG/NCycDB) before running DESeq2, the same category-pooling strategy used generally for de novo (reference-free) metatranscriptome differential expression. The single mismatch, crp, is itself only a modest, non-significant effect in Chuckran et al.'s own reported result, and crp is a global carbon-catabolite regulator rather than a nitrogen-cycle gene per se — a result this

validation reports rather than omits, consistent with the honesty standard applied to MAGMA's, SIGMA's, and PRISM's mixed outcomes above.

Taken together, the five pipelines' real validation results support the paper's central claim: a desktop-native application with zero infrastructure requirements can still reproduce published, ground-truthed multi-omics results end to end, without asking its target user, a wet-lab researcher without dedicated bioinformatics support, to touch a terminal or an HPC scheduler. Every pipeline's real output was checked against an independent, external ground truth (a certified mock-community composition, a reference genome, a published paper's own reported DEGs or fold-changes, or a vendor's own classification of the same read file) rather than against itself, and mixed or partial outcomes (MAGMA's read-based abundance accuracy, SIGMA's PacBio misassemblies, PRISM's undetected Chlamydia trachomatis, DOGMA's single crp mismatch) were reported alongside the strong results rather than smoothed over. As with JOSS and F1000Research's own review policies for research software, this is deliberately not framed as a benchmark against other orchestration platforms (Snakemake, Nextflow, nf-core, Galaxy, anvi'o, KBase); those tools solve a different problem; extensive terminal/HPC-fluent pipeline authoring and execution, at a scale and flexibility OmniOmics does not attempt to match. OmniOmics's contribution is narrower and more specific: five real pipelines, each independently validated, reachable without ever leaving a desktop GUI.

This validation has real limitations, stated plainly rather than left implicit. The pipeline scope is five workflows, not nf-core's 100+ community-maintained pipelines; OmniOmics trades that breadth for depth of integration (one consistent GUI, one plotting environment, one packaging story) across a smaller, curated set. PRISM's PacBio marker-gene path (the vendor's Kinnex 16S mock-community release) remains unvalidated at time of writing — the vendor's own data host was unreachable from both the analysis machine and the HPC cluster used for this work at the time it was set up (connection times out on port 443; DNS itself resolves correctly), a real, external infrastructure constraint documented in the companion repository rather than silently dropped. DOGMA does not attempt long-read-only metatranscriptome assembly, because no validated tool for that specific case exists in the literature at present — a genuine, cited method gap (see the eukrhythmic authors' own multi-assembler rationale, Krinos et al., 2023), not an implementation gap OmniOmics chose not to close. Future work includes additional pipelines, a GROND/full rrn-operon mode for PRISM, and an expanded Plots-tab library for SIGMA.

Acknowledgements

[PENDING]

References

4.1 Platforms and validation datasets

- Arkin AP, Cottingham RW, Henry CS, et al. (2018). KBase: The United States Department of Energy Systems Biology Knowledgebase. Nature Biotechnology, 36(7), 566–569. https://doi.org/10.1038/nbt.4163

- Bolyen E, Rideout JR, Dillon MR, Bokulich NA, et al. (2019). Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nature Biotechnology, 37(8), 852–857. https://doi.org/10.1038/s41587-019-0209-9
- Chuckran PF, Huntemann M, Clum A, et al. (2020). Metagenomes and Metatranscriptomes of a Glucose-Amended Agricultural Soil. Microbiology Resource Announcements, 9(44), e00895-20. https://doi.org/10.1128/MRA.00895-20
- Chuckran PF, Fofanov V, Hungate BA, Morrissey EM, Schwartz E, Walkup J, Dijkstra P (2021). Rapid Response of Nitrogen Cycling Gene Transcription to Labile Carbon Amendments in a Soil Microbial Community. mSystems, 6(3), e00161-21. https://doi.org/10.1128/mSystems.00161-21
- Di Leo D, Nilsson E, Krinos A, Pinhassi J, Lundin D (2025). The Nextflow nf-core/metatdenovo pipeline for reproducible annotation of metatranscriptomes, and more. PeerJ, 13, e20328. https://doi.org/10.7717/peerj.20328
- Eren AM, Kiefl E, Shaiber A, et al. (2021). Community-led, integrated, reproducible multi-omics with anvi'o. Nature Microbiology, 6(1), 3–6. https://doi.org/10.1038/s41564-020-00834-3
- Ewels PA, Peltzer A, Fillinger S, et al. (2020). The nf-core framework for community-curated bioinformatics pipelines. Nature Biotechnology, 38(3), 276–278. https://doi.org/10.1038/s41587-020-0439-x
- Galaxy Community (2022). The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2022 update. Nucleic Acids Research, 50(W1), W345–W351. https://doi.org/10.1093/nar/gkac247
- Krakau S, Straub D, Gourlé H, Gabernet G, Nahnsen S (2022). nf-core/mag: a best-practice pipeline for metagenome hybrid assembly and binning. NAR Genomics and Bioinformatics, 4(1), lqac007. https://doi.org/10.1093/nargab/lqac007
- Krinos AI, Cohen NR, Follows MJ, Alexander H (2023). Reverse engineering environmental metatranscriptomes clarifies best practices for eukaryotic assembly. BMC Bioinformatics, 24(1), 74. https://doi.org/10.1186/s12859-022-05121-y
- McIntyre ABR, Alexander N, Grigorev K, Bezdan D, Sichtig H, Chiu CY, Mason CE (2019). Single-molecule sequencing detection of N6-methyladenine in microbial reference materials. Nature Communications, 10, 579. https://doi.org/10.1038/s41467-019-08289-9
- Di Tommaso P, Chatzou M, Floden EW, Barja PP, Palumbo E, Notredame C (2017). Nextflow enables reproducible computational workflows. Nature Biotechnology, 35(4), 316–319. https://doi.org/10.1038/nbt.3820
- Nicholls SM, Quick JC, Tang S, Loman NJ (2019). Ultra-deep, long-read nanopore sequencing of mock microbial community standards. GigaScience, 8(5), giz043. https://doi.org/10.1093/gigascience/giz043
- Oxford Nanopore Technologies (2025). Zymo 16S mock community open data release (zymo_16s_2025.09). EPI2ME blog. https://epi2me.nanoporetech.com/zymo_16s_2025.09/
- Mölder F, Jablonski KP, Letcher B, et al. (2021). Sustainable data analysis with Snakemake. F1000Research, 10, 33. https://doi.org/10.12688/f1000research.29032.3
- Petit RA III, Read TD (2020). Bactopia: a flexible pipeline for complete analysis of bacterial genomes. mSystems, 5(4), e00190-20. https://doi.org/10.1128/mSystems.00190-20
- Stamouli S, Beber ME, Normark T, Christensen II TA, Andersson-Li L, Borry M, Jamy M, nf-core community, Fellows Yates JA (2023). nf-core/taxprofiler: Highly parallelised and flexible pipeline for metagenomic taxonomic classification and profiling. bioRxiv preprint. https://doi.org/10.1101/2023.10.20.563221

- Varet H, Brillet-Guéguen L, Coppée JY, Dillies MA (2016). SARTools: A DESeq2- and edgeR-Based R Pipeline for Comprehensive Differential Analysis of RNA-Seq Data. PLoS ONE, 11(6), e0157022. https://doi.org/10.1371/journal.pone.0157022
- Wintenberg M, Manglass L, Martinez NE, Blenner M (2023). Global Transcriptional Response of Escherichia coli Exposed In Situ to Different Low-Dose Ionizing Radiation Sources. mSystems, 8(2), e00718-22. https://doi.org/10.1128/msystems.00718-22

4.2 Tools and reference databases used by the five pipelines

OmniOmics packages, but does not modify, the following third-party tools and reference databases across its five pipelines. Listed for completeness and correct attribution; extracted directly from each pipeline's own manifest (pipeline.yaml), not independently curated for this list, so it reflects exactly what a user's real run actually invokes.

4.2.1 Tools

- Alonge M, Lebeigle L, Kirsche M, Jenike K, Ou S, Aganezov S, Wang X, Lippman ZB, Schatz MC, Soyk S. Automated assembly scaffolding using RagTag elevates a new tomato system for high-throughput genome editing. Genome Biology. 2022;23:258. https://doi.org/10.1186/s13059-022-02823-7
- Antipov D, Korobeynikov A, McLean JS, Pevzner PA. hybridSPAdes: an algorithm for hybrid assembly of short and long reads. Bioinformatics. 2016;32(7):1009-1015. https://doi.org/10.1093/bioinformatics/btv688
- Aroney STN, Newell RJP, Nissen JN, Camargo AP, Tyson GW, Woodcroft BJ. CoverM: read alignment statistics for metagenomics. Bioinformatics. 2025;41(4):btaf147. https://doi.org/10.1093/bioinformatics/btaf147
- Bland C, Ramsey TL, Sabree F, Lowe M, Brown K, Kyrpides NC, Hugenholtz P. CRISPR Recognition Tool (CRT): a tool for automatic detection of clustered regularly interspaced palindromic repeats. BMC Bioinformatics. 2007;8:209. https://doi.org/10.1186/1471-2105-8-209
- Blin K, Shaw S, Augustijn HE, et al. antiSMASH 7.0: new and improved predictions for detection, regulation, chemical structures and visualisation. Nucleic Acids Research. 2023;51(W1):W46-W50. https://doi.org/10.1093/nar/gkad344
- Buchfink B, Reuter K, Drost HG. Sensitive protein alignments at tree-of-life scale using DIAMOND. Nature Methods. 2021;18(4):366-368. https://doi.org/10.1038/s41592-021-01101-x
- Buchfink B, Xie C, Huson DH. Fast and sensitive protein alignment using DIAMOND. Nature Methods. 2015;12(1):59-60. https://doi.org/10.1038/nmeth.3176
- Bushmanova E, Antipov D, Lapidus A, Prjibelski AD. rnaSPAdes: a de novo transcriptome assembler and its application to RNA-Seq data. GigaScience. 2019;8(9):giz100. https://doi.org/10.1093/gigascience/giz100
- Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJ, Holmes SP. DADA2: High-resolution sample inference from Illumina amplicon data. Nature Methods. 2016;13(7):581-583. https://doi.org/10.1038/nmeth.3869
- Callahan BJ, Wong J, Siavoshi C, Vuong C, Sørensen EA, Ravel J, Knight R. High-throughput amplicon sequencing of the full-length 16S rRNA gene with single-nucleotide resolution. Nucleic Acids Research. 2019;47(18):e103. https://doi.org/10.1093/nar/gkz569
- Camargo AP, Roux S, Schulz F, et al. Identification of mobile genetic elements with geNomad. Nature Biotechnology. 2024;42:1303-1312. https://doi.org/10.1038/s41587-023-01953-y
- Cantalapiedra CP, Hernandez-Plaza A, Letunic I, Bork P, Huerta-Cepas J. eggNOG-mapper v2: functional annotation, orthology assignments, and domain prediction at the metagenomic scale. Molecular Biology and Evolution. 2021;38(12):5825-5829. https://doi.org/10.1093/molbev/msab293
- Chan PP, Lin BY, Mak AJ, Lowe TM. tRNAscan-SE 2.0: improved detection and functional classification of transfer RNA genes. Nucleic Acids Research. 2021;49(16):9077-9096. https://doi.org/10.1093/nar/gkab688
- Chaumeil PA, Mussig AJ, Hugenholtz P, Parks DH. GTDB-Tk v2: memory friendly classification with the genome taxonomy database. Bioinformatics. 2022;38(23):5315-5316. https://doi.org/10.1093/bioinformatics/btac672

- Chaumeil PA, Mussig AJ, Hugenholtz P, Parks DH. GTDB-Tk: a toolkit to classify genomes with the Genome Taxonomy Database. Bioinformatics. 2020;36(6):1925-1927. https://doi.org/10.1093/bioinformatics/btz848
- Chen S, Zhou Y, Chen Y, Gu J. fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics. 2018;34(17):i884-i890. https://doi.org/10.1093/bioinformatics/bty560
- Chklovski A, Parks DH, Woodcroft BJ, Tyson GW. CheckM2: a rapid, scalable and accurate tool for assessing microbial genome quality using machine learning. Nature Methods. 2023;20:1203-1212. https://doi.org/10.1038/s41592-023-01940-w
- Curry KD, Wang Q, Nute MG, et al. Emu: species-level microbial community profiling of full-length 16S rRNA Oxford Nanopore sequencing data. Nature Methods. 2022;19:845-853. https://doi.org/10.1038/s41592-022-01520-4
- Danecek P, Bonfield JK, Liddle J, Marshall J, Ohan V, Pollard MO, Whitwham A, Keane T, McCarthy SA, Davies RM, Li H. Twelve years of SAMtools and BCFtools. GigaScience. 2021;10(2):giab008. https://doi.org/10.1093/gigascience/giab008
- De Coster W, Rademakers R. NanoPack2: population-scale evaluation of long-read sequencing data. Bioinformatics. 2023;39(5):btad311. https://doi.org/10.1093/bioinformatics/btad311
- Deng ZL, Münch PC, Mreches R, McHardy AC. Rapid and accurate identification of ribosomal RNA sequences via deep learning. Nucleic Acids Research. 2022;50(10):e60. https://doi.org/10.1093/nar/gkac112
- Douglas GM, Maffei VJ, Zaneveld JR, et al. PICRUSt2 for prediction of metagenome functions. Nature Biotechnology. 2020;38:685-688. https://doi.org/10.1038/s41587-020-0548-6
- Ewels P, Magnusson M, Lundin S, Käller M. MultiQC: summarize analysis results for multiple tools and samples in a single report. Bioinformatics. 2016;32(19):3047-3048. https://doi.org/10.1093/bioinformatics/btw354
- Feldgarden M, Brover V, Gonzalez-Escalona N, et al. AMRFinderPlus and the Reference Gene Catalog facilitate examination of the genomic links among antimicrobial resistance, stress response, and virulence. Scientific Reports. 2021;11:12728. https://doi.org/10.1038/s41598-021-91456-0
- Garber AI, Nealson KH, Okamoto A, McAllister SM, Chan CS, Barco RA, Merino N. FeGenie: A Comprehensive Tool for the Identification of Iron Genes and Iron Gene Neighborhoods in Genome and Metagenome Assemblies. Frontiers in Microbiology. 2020;11:37. https://doi.org/10.3389/fmicb.2020.00037
- Grabherr MG, Haas BJ, Yassour M, et al. Full-length transcriptome assembly from RNA-seq data without a reference genome. Nature Biotechnology. 2011;29(7):644-652. https://doi.org/10.1038/nbt.1883
- Guo J, Bolduc B, Zayed AA, et al. VirSorter2: a multi-classifier, expert-guided approach to detect diverse DNA and RNA viruses. Microbiome. 2021;9:37. https://doi.org/10.1186/s40168-020-00990-y
- Gurevich A, Saveliev V, Vyahhi N, Tesler G. QUAST: quality assessment tool for genome assemblies. Bioinformatics. 2013;29(8):1072-1075. https://doi.org/10.1093/bioinformatics/btt086
- Haas BJ, Papanicolaou A, Yassour M, et al. De novo transcript sequence reconstruction from RNA-seq using the Trinity platform for reference generation and analysis. Nature Protocols. 2013;8(8):1494-1512. https://doi.org/10.1038/nprot.2013.084
- Hakimzadeh A, Mikryukov V, Metsoja M, Tedersoo L, Anslan S. Are we throwing away good data? Evaluation of chimera detection algorithms on long-read amplicons reveals high false-positive rates across algorithms. PeerJ. 2025. https://doi.org/10.7717/peerj.20456
- Huerta-Cepas J, Szklarczyk D, Heller D, et al. eggNOG 5.0: a hierarchical, functionally and phylogenetically annotated orthology resource based on 5090 organisms and 2502 viruses. Nucleic Acids Research. 2019;47(D1):D309-D314. https://doi.org/10.1093/nar/gky1085
- Hyatt D, Chen GL, LoCascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. BMC Bioinformatics. 2010;11:119. https://doi.org/10.1186/1471-2105-11-119
- Jolley KA, Maiden MCJ. BIGSdb: scalable analysis of bacterial genome variation at the population level. BMC Bioinformatics. 2010;11:595. https://doi.org/10.1186/1471-2105-11-595
- Kang DD, Li F, Kirton E, Thomas A, Egan R, An H, Wang Z. MetaBAT 2: an adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies. PeerJ. 2019;7:e7359. https://doi.org/10.7717/peerj.7359

- Karlicki M, Antonowicz S, Karnkowska A. Tiara: deep learning-based classification system for eukaryotic sequences. Bioinformatics. 2022;38(2):344-350. https://doi.org/10.1093/bioinformatics/btab672
- Khot V, Zorz J, Gittins DA, et al. CANT-HYD: A Curated Database of Phylogeny-Derived Hidden Markov Models for Annotation of Marker Genes Involved in Hydrocarbon Degradation. Frontiers in Microbiology. 2021;12:764058. https://doi.org/10.3389/fmicb.2021.764058
- Kieft K, Zhou Z, Anantharaman K. VIBRANT: automated recovery, annotation and curation of microbial viruses, and evaluation of viral community function from genomic sequences. Microbiome. 2020;8:90. https://doi.org/10.1186/s40168-020-00867-0
- Kolmogorov M, Yuan J, Lin Y, Pevzner PA. Assembly of long, error-prone reads using repeat graphs. Nature Biotechnology. 2019;37:540-546. https://doi.org/10.1038/s41587-019-0072-8
- Krinos AI, Hu SK, Cohen NR, Alexander H. EUKulele: Taxonomic annotation of the unsung eukaryotic microbes. Journal of Open Source Software. 2021;6(57):2817. https://doi.org/10.21105/joss.02817
- Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. Nature Methods. 2012;9(4):357-359. https://doi.org/10.1038/nmeth.1923
- Larralde M. Pyrodigal: Python bindings and interface to Prodigal, an efficient method for gene prediction in prokaryotes. Journal of Open Source Software. 2022;7(72):4296. https://doi.org/10.21105/joss.04296
- Li D, Liu CM, Luo R, Sadakane K, Lam TW. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. Bioinformatics. 2015;31(10):1674-1676. https://doi.org/10.1093/bioinformatics/btv033
- Li H. Minimap2: pairwise alignment for nucleotide sequences. Bioinformatics. 2018;34(18):3094-3100. https://doi.org/10.1093/bioinformatics/bty191
- Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics. 2014;30(7):923-930. https://doi.org/10.1093/bioinformatics/btt656
- Lu J, Breitwieser FP, Thielen P, Salzberg SL. Bracken: estimating species abundance in metagenomics data. PeerJ Computer Science. 2017;3:e104. https://doi.org/10.7717/peerj-cs.104
- Lu J, Rincon N, Wood DE, Breitwieser FP, Pockrandt C, Langmead B, Salzberg SL, Steinegger M. Metagenome analysis using the Kraken software suite. Nature Protocols. 2022;17:2815-2839. https://doi.org/10.1038/s41596-022-00738-y
- Mainguy J, Hoede C. Binette: a fast and accurate bin refinement tool to construct high quality Metagenome Assembled Genomes. Journal of Open Source Software. 2024;9(102):6782. https://doi.org/10.21105/joss.06782
- Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. EMBnet.journal. 2011;17(1):10-12. https://doi.org/10.14806/ej.17.1.200
- Menzel P, Ng KL, Krogh A. Fast and sensitive taxonomic classification for metagenomics with Kaiju. Nature Communications. 2016;7:11257. https://doi.org/10.1038/ncomms11257
- Mikheenko A, Saveliev V, Gurevich A. MetaQUAST: evaluation of metagenome assemblies. Bioinformatics. 2016;32(7):1088-1090. https://doi.org/10.1093/bioinformatics/btv697
- Navarro-Muñoz JC, Selem-Mojica N, Mullowney MW, et al. A computational framework to explore large-scale biosynthetic diversity. Nature Chemical Biology. 2020;16(1):60-68. https://doi.org/10.1038/s41589-019-0400-9
- Nayfach S, Camargo AP, Schulz F, Eloe-Fadrosh E, Roux S, Kyrpides NC. CheckV assesses the quality and completeness of metagenome-assembled viral genomes. Nature Biotechnology. 2021;39:578-585. https://doi.org/10.1038/s41587-020-00774-7
- Nissen JN, Johansen J, Allesøe RL, Sønderby CK, Armenteros JJA, Grønbech CH, Jensen LJ, Nielsen HB, Petersen TN, Winther O, Rasmussen S. Improved metagenome binning and assembly using deep variational autoencoders. Nature Biotechnology. 2021;39:555-560. https://doi.org/10.1038/s41587-020-00777-4
- Nurk S, Meleshko D, Korobeynikov A, Pevzner PA. metaSPAdes: a new versatile metagenomic assembler. Genome Research. 2017;27(5):824-834. https://doi.org/10.1101/gr.213959.116
- Olm MR, Brown CT, Brooks B, Banfield JF. dRep: a tool for fast and accurate genomic comparisons that enables improved genome recovery from metagenomes through de-replication. The ISME Journal. 2017;11(12):2864-2868. https://doi.org/10.1038/ismej.2017.126

- Orakov A, Fullam A, Coelho LP, Khedkar S, Szklarczyk D, Mende DR, Schmidt TSB, Bork P. GUNC: detection of chimerism and contamination in prokaryotic genomes. Genome Biology. 2021;22:178. https://doi.org/10.1186/s13059-021-02393-0
- Pan S, Zhao XM, Coelho LP. SemiBin2: self-supervised contrastive learning leads to better MAGs for short- and long-read sequencing. Bioinformatics. 2023;39(Suppl 1):i21-i29. https://doi.org/10.1093/bioinformatics/btad209
- Patro R, Duggal G, Love MI, Irizarry RA, Kingsford C. Salmon provides fast and bias-aware quantification of transcript expression. Nature Methods. 2017;14(4):417-419. https://doi.org/10.1038/nmeth.4197
- Qian L, Yu X, Zhou J, et al. MCycDB: A curated database for comprehensively profiling methane cycling processes of environmental microbiomes. Molecular Ecology Resources. 2022;22(5):1803-1823. https://doi.org/10.1111/1755-0998.13589
- Robertson J, Bessonov K, Schonfeld J, Nash JHE. Universal whole-sequence-based plasmid typing and its utility to prediction of host range and epidemiological surveillance. Microbial Genomics. 2020;6(10):mgen000435. https://doi.org/10.1099/mgen.0.000435
- Robertson J, Nash JHE. MOB-suite: software tools for clustering, reconstruction and typing of plasmids from draft assemblies. Microbial Genomics. 2018;4(8):e000206. https://doi.org/10.1099/mgen.0.000206
- Rognes T, Flouri T, Nichols B, Quince C, Mahé F. VSEARCH: a versatile open source tool for metagenomics. PeerJ. 2016;4:e2584. https://doi.org/10.7717/peerj.2584
- Schwengers O, Jelonek L, Dieckmann MA, Beyvers S, Blom J, Goesmann A. Bakta: rapid and standardized annotation of bacterial genomes via alignment-free sequence identification. Microbial Genomics. 2021;7(11):000685. https://doi.org/10.1099/mgen.0.000685
- Steinegger M, Söding J. Clustering huge protein sequence sets in linear time. Nature Communications. 2018;9:2542. https://doi.org/10.1038/s41467-018-04964-5
- Steinegger M, Söding J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology. 2017;35(11):1026-1028. https://doi.org/10.1038/nbt.3988
- Tu Q, Lin L, Cheng L, Deng Y, He Z. NCycDB: a curated integrative database for fast and accurate metagenomic profiling of nitrogen cycling genes. Bioinformatics. 2019;35(6):1040-1048. https://doi.org/10.1093/bioinformatics/bty741
- Vaser R, Sović I, Nagarajan N, Šikić M. Fast and accurate de novo genome assembly from long uncorrected reads. Genome Research. 2017;27(5):737-746. https://doi.org/10.1101/gr.214270.116
- Wang L, Wang S, Li W. RSeQC: quality control of RNA-seq experiments. Bioinformatics. 2012;28(16):2184-2185. https://doi.org/10.1093/bioinformatics/bts356
- Wick RR, Judd LM, Gorrie CL, Holt KE. Unicycler: resolving bacterial genome assemblies from short and long sequencing reads. PLOS Computational Biology. 2017;13(6):e1005595. https://doi.org/10.1371/journal.pcbi.1005595
- Wood DE, Lu J, Langmead B. Improved metagenomic analysis with Kraken 2. Genome Biology. 2019;20:257. https://doi.org/10.1186/s13059-019-1891-0
- Wu YW, Simmons BA, Singer SW. MaxBin 2.0: an automated binning algorithm to recover genomes from multiple metagenomic datasets. Bioinformatics. 2016;32(4):605-607. https://doi.org/10.1093/bioinformatics/btv638
- Xie Z, Tang H. ISEScan: automated identification of insertion sequence elements in prokaryotic genomes. Bioinformatics. 2017;33(21):3340-3347. https://doi.org/10.1093/bioinformatics/btx433
- Yu X, Zhou J, Song W, Xu M, He Q, Peng Y, Tian Y, Wang C, Shu L, Wang S, Yan Q, Liu J, He Z. SCycDB: A curated functional gene database for metagenomic profiling of sulphur cycling pathways. Molecular Ecology Resources. 2021;21(3):924-940. https://doi.org/10.1111/1755-0998.13306
- Zheng J, Ge Q, Yan Y, Zhang X, Huang L, Yin Y. dbCAN3: automated carbohydrate-active enzyme and substrate annotation. Nucleic Acids Research. 2023;51(W1):W115-W121. https://doi.org/10.1093/nar/gkad328
- Zhou S, Liu B, Zheng D, Chen L, Yang J. VFDB 2025: an integrated resource for exploring anti-virulence compounds. Nucleic Acids Research. 2025;53(D1):D871-D877. https://doi.org/10.1093/nar/gkae968
- von Meijenfeldt FAB, Arkhipova K, Cambuy DD, Coutinho FH, Dutilh BE. Robust taxonomic classification of uncharted microbial sequences and bins with CAT and BAT. Genome Biology. 2019;20:217. https://doi.org/10.1186/s13059-019-1817-x

4.2.2 Reference databases

- Abarenkov K, Zirk A, Piirmann T, et al. UNITE general FASTA release for Fungi. Version 10.0. https://doi.org/10.15156/BIO/2959332
- Alishum A. DADA2 formatted 16S rRNA gene sequences for both bacteria & archaea (GTDB release 220). Zenodo. https://doi.org/10.5281/zenodo.13984843
- Benitez-Paez A, Hartkopf F, Muth T, et al. GROND: a quality-checked and publicly available database of full-length 16S-ITS-23S rRNA operon sequences. Zenodo. https://doi.org/10.5281/zenodo.10889037
- Cusco A, Catozzi C, Vines J, Sanchez A, Francino O. Microbiota profiling with long amplicons using Nanopore sequencing: full-length 16S rRNA gene and the 16S-ITS-23S of the rrn operon. F1000Research. 2019;7:1755. https://doi.org/10.12688/f1000research.16817.2
- Fullam A, Letunic I, Schmidt TSB, et al. proGenomes3: approaching one million accurately and consistently annotated high-quality prokaryotic genomes. Nucleic Acids Research. 2023;51(D1):D760-D766. https://doi.org/10.1093/nar/gkac1078
- Iwasaki W, Fukunaga T, Isagozawa R, et al. MitoFish and MitoAnnotator: a mitochondrial genome database of fish with an accurate and automatic annotation pipeline. Molecular Biology and Evolution. 2013;30(11):2531-2540. https://doi.org/10.1093/molbev/mst141
- Keeling PJ, Burki F, Wilcox HM, et al. The Marine Microbial Eukaryote Transcriptome Sequencing Project (MMETSP): Illuminating the Functional Diversity of Eukaryotic Life in the Oceans through Transcriptome Sequencing. PLoS Biology. 2014;12(6):e1001889. https://doi.org/10.1371/journal.pbio.1001889
- Klemetsen T, Raknes IA, Fu J, et al. The MAR databases: development and implementation of databases specific for marine metagenomics. Nucleic Acids Research. 2018;46(D1):D692-D699. https://doi.org/10.1093/nar/gkx1036
- Leray M, Knowlton N, Machida RJ. MIDORI2: A collection of quality controlled, preformatted, and regularly updated reference databases for taxonomic assignment of eukaryotic mitochondrial sequences. Environmental DNA. 2022;4(4):894-907. https://doi.org/10.1002/edn3.303
- Lim J, Thompson LR. Mitohelper: A mitochondrial reference sequence analysis tool for fish eDNA studies. Environmental DNA. 2021;3(6):1053-1060. https://doi.org/10.1002/edn3.187
- Mende DR, Letunic I, Maistrenko OM, et al. proGenomes2: an improved database for accurate and consistent habitat, taxonomic and functional annotations of prokaryotic genomes. Nucleic Acids Research. 2020;48(D1):D621-D625. https://doi.org/10.1093/nar/gkz1002
- Miya M, Sato Y, Fukunaga T, et al. MiFish, a set of universal PCR primers for metabarcoding environmental DNA from fishes: detection of more than 230 subtropical marine species. Royal Society Open Science. 2015;2(7):150088. https://doi.org/10.1098/rsos.150088
- Nilsson RH, Larsson KH, Taylor AFS, et al. The UNITE database for molecular identification of fungi: handling dark taxa and parallel taxonomic classifications. Nucleic Acids Research. 2019;47(D1):D259-D264. https://doi.org/10.1093/nar/gky1022
- Parks DH, Chuvochina M, Chaumeil PA, Rinke C, Mussig AJ, Hugenholtz P. A complete domain-to-species taxonomy for Bacteria and Archaea. Nature Biotechnology. 2020;38:1079-1086. https://doi.org/10.1038/s41587-020-0501-8
- Parks DH, Chuvochina M, Rinke C, Mussig AJ, Chaumeil PA, Hugenholtz P. GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. Nucleic Acids Research. 2022;50(D1):D785-D794. https://doi.org/10.1093/nar/gkab776
- Parks DH, Chuvochina M, Waite DW, et al. A standardized bacterial taxonomy based on genome phylogeny substantially revises the tree of life. Nature Biotechnology. 2018;36:996-1004. https://doi.org/10.1038/nbt.4229
- Paysan-Lafosse T, Andreeva A, Blum M, et al. The Pfam protein families database: embracing AI/ML. Nucleic Acids Research. 2025;53(D1):D523-D534. https://doi.org/10.1093/nar/gkae997
- Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P, Peplies J, Glöckner FO. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. Nucleic Acids Research. 2013;41(D1):D590-D596. https://doi.org/10.1093/nar/gks1219

- Sato Y, Miya M, Fukunaga T, Sado T, Iwasaki W. MitoFish and MiFish Pipeline: A Mitochondrial Genome Database of Fish with an Analysis Pipeline for Environmental DNA Metabarcoding. Molecular Biology and Evolution. 2018;35(6):1553-1555. https://doi.org/10.1093/molbev/msy074

Data Accessibility and Benefit-Sharing

OmniOmics's source code is available at github.com/oliv-angelus/OmniOmics (MIT license), including a packaged Debian build (.deb). The exact configuration, curated ground truth, and real observed-vs-expected validation results described in this manuscript are available at github.com/oliv-angelus/OmniOmics-validation (MIT for scripts/configs, CC-BY-4.0 for data). A permanent, versioned Zenodo DOI for both repositories **[PENDING: to be minted before submission].**

All datasets reanalyzed for validation are already public, deposited by their original authors or by the sequencing-platform vendor (SRA/ENA/GEO/ONT open-data accessions, cited per pipeline in Table 2 and in the companion repository). No new biological samples were collected for this work; the Benefit-Sharing provisions of the Nagoya Protocol / Convention on Biological Diversity are therefore not applicable.

Author Contributions

[PENDING]

Tables and Figures

[PENDING: Table 3 (real observed-vs-expected results) should be expanded into the full per-metric version once every pipeline run is complete. Figures: architecture diagram (client/backend/pipeline-registry), one assembly-quality figure (e.g., SIGMA's short-only vs. hybrid contig-count/genome-fraction comparison) as a visual anchor for the validation argument.]

[PENDING: Figure: per-pipeline workflow diagrams (one panel per pipeline, tool-labeled DAG). Real draw.io SVGs already exist for MAGMA, SIGMA, and DOGMA (pipelines/<id>/<ID>.drawio.svg in the OmniOmics repo); DEFINE and PRISM diagrams are being drawn by the author in the same style; insert all 5 as one multi-panel figure once both are ready, not before (keeps the 5 panels visually consistent).]